THE UNIVERSITY OF HONG KONG
SCHOOL OF COMPUTING AND DATA SCIENCE

COMP9602 Optimization

Project (24%)

Overview

In this project, you are required to model/solve an optimization problem in your research area (or an area you select) and write a report on the related literature study, problem formulation, and solution of the optimization problem.

To complete this project, you should first decide a research area that you are working on in your Ph.D./MPhil study or another area you are interested in, and survey the optimization problems in the area, which have been discussed in the existing literature.

Based on your literature study, you will identify a new optimization problem that (1) interests you, (2) is important in the topic area, and (3) has not been solved by any existing literature. You will then model the problem into a (convex) program.

Next, you will design an algorithm to solve the problem that you have formulated, based on algorithms we learn in class, or any algorithm in literature you have studied on your own. It is desirable that you also analyze the convergence properties and complexity of the algorithm you design.

You should then implement your algorithm using Python or Matlab, and test it with typical inputs from the topic area, with respect to
(1) correctness of the algorithm, i.e., whether it can find the global optimal solution.
(2) performance of the algorithm, e.g., in terms of the number of iterations it takes to converge to the optimum.

Finally, all the above should be reported in the project report, with detailed requirements in the section of "Requirements on Project Report".

More on Topic Selection

For those whose research is related to optimization, you are strongly encouraged to work on one problem in your research area. You may consult your supervisor on a related topic, such that the study you carry out in this project can potentially lead to a paper submission.

Requirements on Project Report

Your report should follow the IEEE double-column 10-pt font format, with a page limit of 5.

A maximum number of 2 students are allowed to form a group to work on the same report. Individual project is encouraged (please refer to the bonus scheme listed in the "Grading" section).

The report should include the following sections:

1. Title
   Specify the area and the problem you are formulating.

2. Abstract
   Briefly state: the current status and the significance of the problem; the challenge to solve the problem; what you will do in this project on modeling/solving the problem; the performance of your solution algorithm.

3. Introduction
   A more elaborated description of the items included in the abstract.

4. Related work
   Include your survey of the existing literature on the optimization problems in the topic area. You should discuss the optimization problems that have been studied before, the approaches that have been designed/applied to solve the problems, and the pros and cons of these optimization models/algorithms.

5. Model
   Model the (convex) program in this section. You should give detailed definitions for the notation used, and explain the rationale and implications behind the formulation of the objective function and each constraint.

6. Algorithm
   Describe the detailed algorithm you have designed to solve the program, desirably together with analysis of the convergence properties and complexity of the algorithm.

7. Evaluation
   Present the evaluation results of the algorithm based on your Python or Matlab implementation. Include figures plotted with Python or Matlab, which show the convergence/performance of the algorithm under different input settings.

8. Conclusion
   A brief summary of what you have done in the project, key observations from your algorithm evaluation, and any future directions to explore.

9. Reference
   You should list all the technical articles/papers/websites you referred to in this project. You should number each of the reference items, and cite them using their index in the main text.

Project Schedule

Deadline	Task	Means of submission
23:59 Sunday, November 15, 2026	Topic selection and group formation	Email your topic selection and group member info** to the TA (ml621@connect.hku.hk) and cc the professor (cwu@cs.hku.hk) [note that only one email needs to be sent for each group]
23:59 Tuesday, December 22, 2026	Project report due	Please submit the report PDF on Moodle

**Your email should include: (1) the members in your group (name + department/university + HKU student number); (2) tentative title of your project report; (3) a brief description (with a few sentences) of the research area and the optimization problem you plan to investigate.

Grading (on a 100% basis)

Email on topic selection and group formulation: 5%
Thoroughness and classification in literature survey: 15%
Correctness and clear explanation of problem modeling: 15%
Correctness and efficiency of algorithm design: 20%
Algorithm evaluation: 15%
Quality of writing (English presentation, clarity, figures, tables): 15%
Completeness of the report (abstract, introduction, conclusion, reference): 15%

Bonus scheme for individual projects:
1 student in a group: your mark + 10 (upper-bounded by 100)

Note: copying the model/algorithm from any existing paper instead of developing your own will be treated as a serious plagiarism.